MINISTÉRIO DA CIÊNCIA E TECNOLOGIA
INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS

Planning your research

Research methods course

Gilberto Camara

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How to finish a thesis?

You'll need four qualities...

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How to finish a thesis?

You'll need four qualities...

Discipline...

... Discipline

... More discipline

... and physical and mental health

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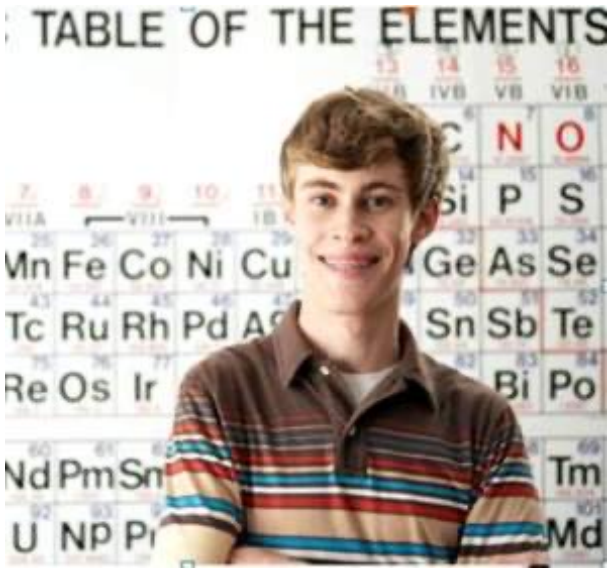
Research Phases

Advisor's Choice

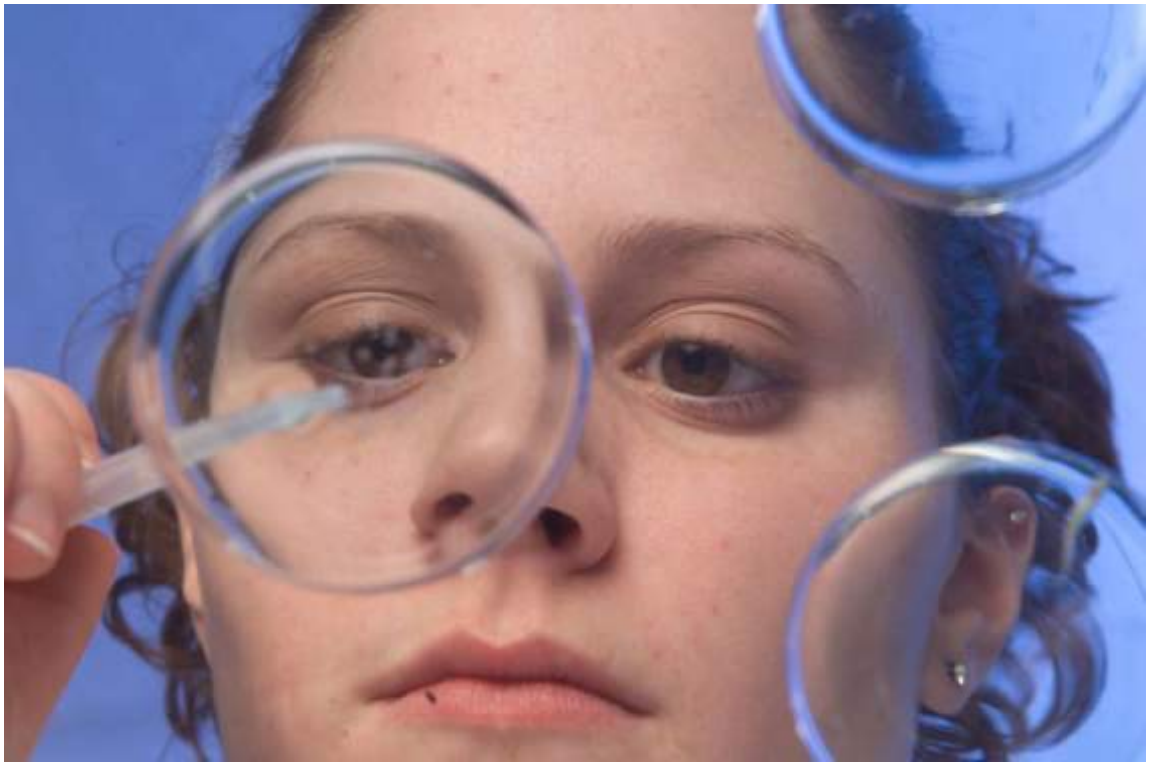


Problem Definition

Conducting the Research

Publication and defense

How to define a problem?

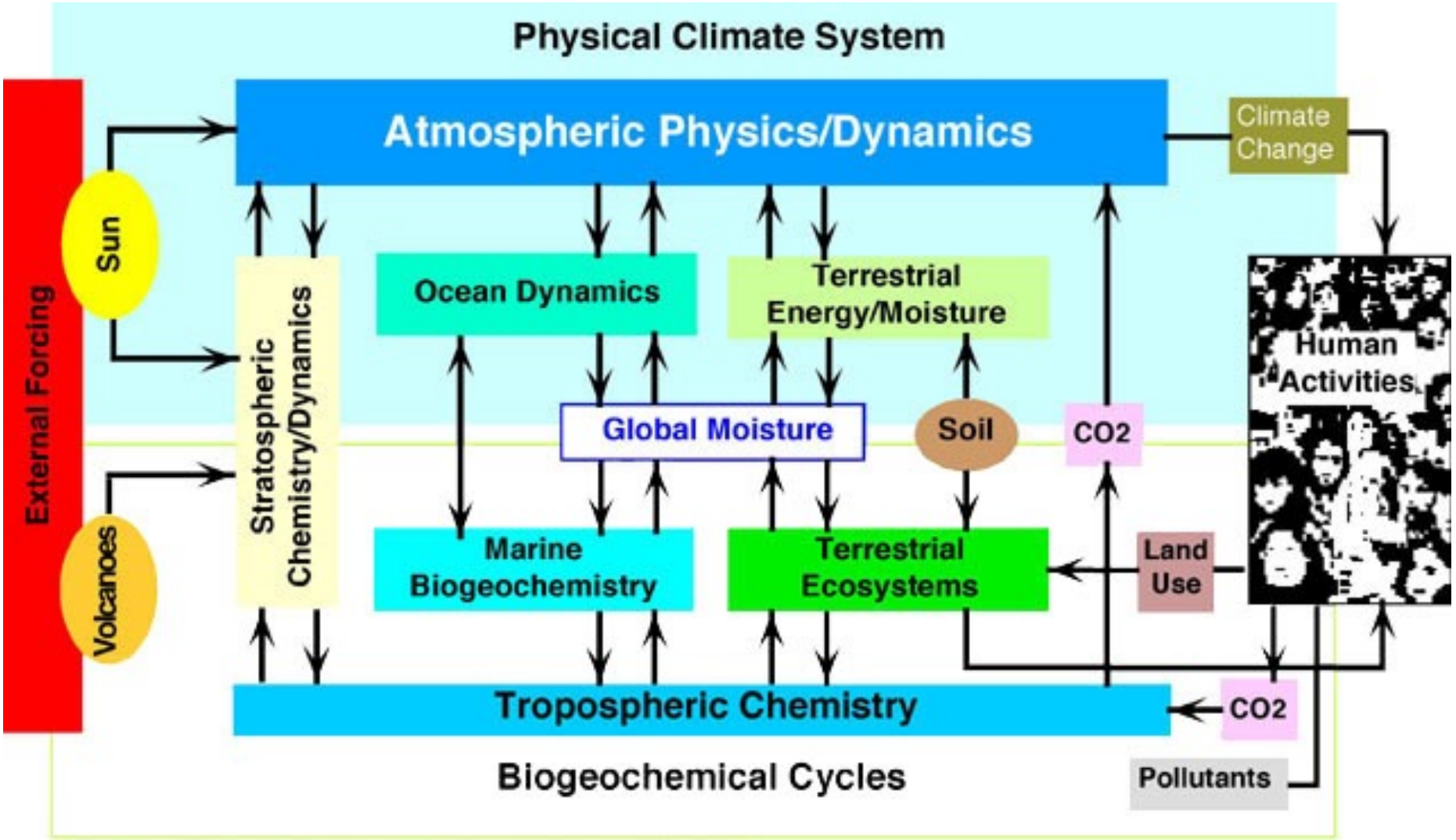


Choose questions from a research programme

Know the state-of-the-art (literature)

Be part of informal collaboration networks

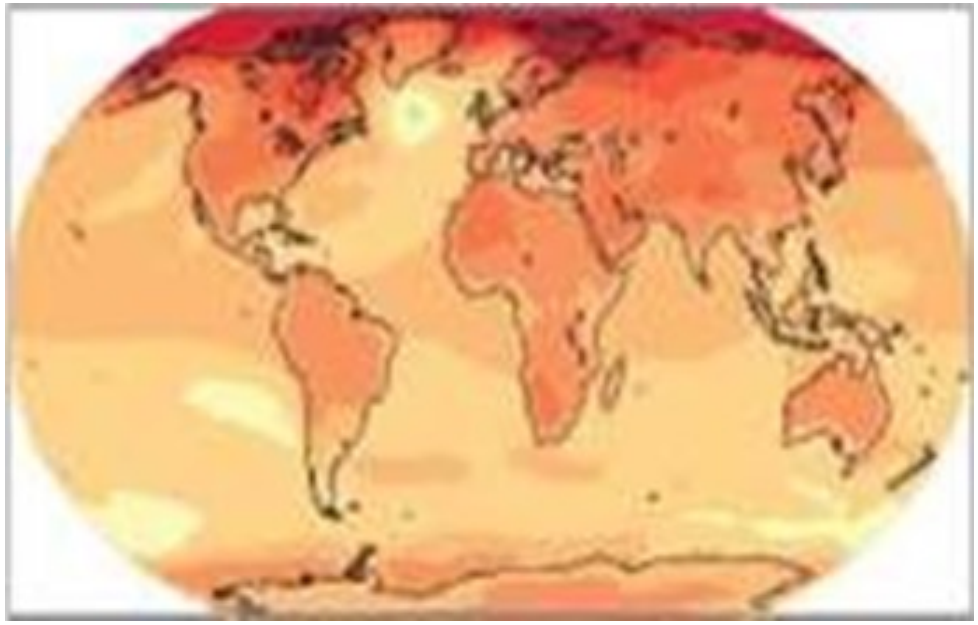
Earth System Science



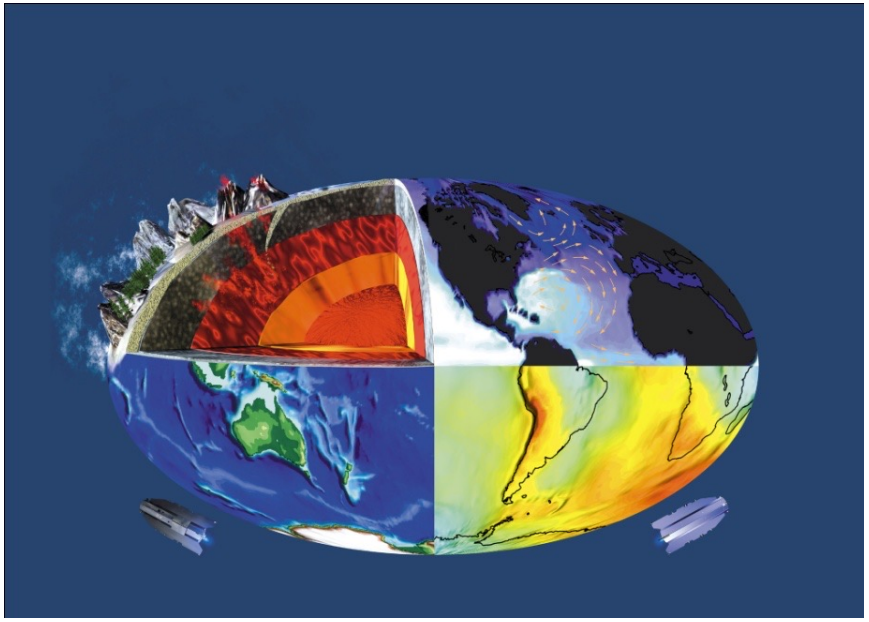
What is a problem in Earth System Science?

How to reduce the consumption of natural resources by humanity?

What will happen to the planet if the concentration of GHG in the atmosphere is 750 ppm CO2eq?



What is a problem in Earth System Science?



How much GHG emissions will Brazil produce in 2030, given the current consumption pattern and the expected growth of the population and the economy?

What are the personal incentives to reduce individual consumption?

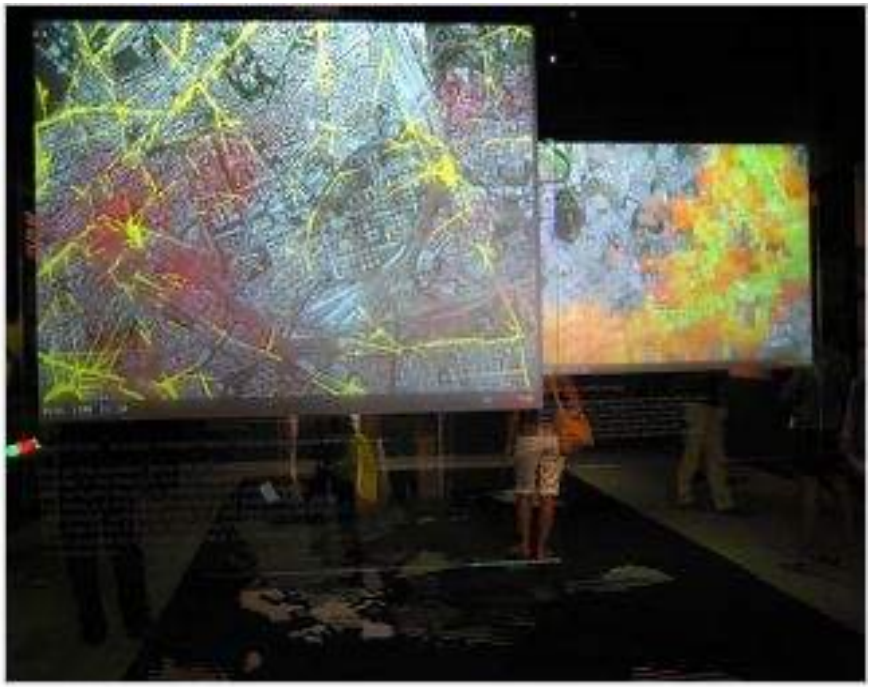
What is a problem in Earth System Science?

How can global environmental changes affect food security?

What adaptations in the livelihoods in the Caatinga will be necessary in an A2-IPCC scenario?

What is a problem in Computer Science?

Computation is synthetic in the sense that many of the phenomena computer scientists and engineers study are created by humans rather than occurring naturally in the physical world.



What is a problem in Computer Science?

When one discovers a fact about nature, it is a contribution per se, no matter how small. Since anyone can create something new [in a synthetic field], that alone does not establish a contribution. Rather, one must show that the creation is “better” (Fred Brooks)

What is a problem in Computer Science?

"Better" can mean many things:

"solves a problem in less time,"
"solves a larger class of problems,"
"is more efficient of resources,"
"is more expressive by some criterion,"
"is more visually appealing,"
"presents a totally new capability," etc.

Required content for Computer Science papers

What exactly is the problem being solved?

Are the ideas in the paper new? Can you state them concisely?

How is your work significantly different from related works?

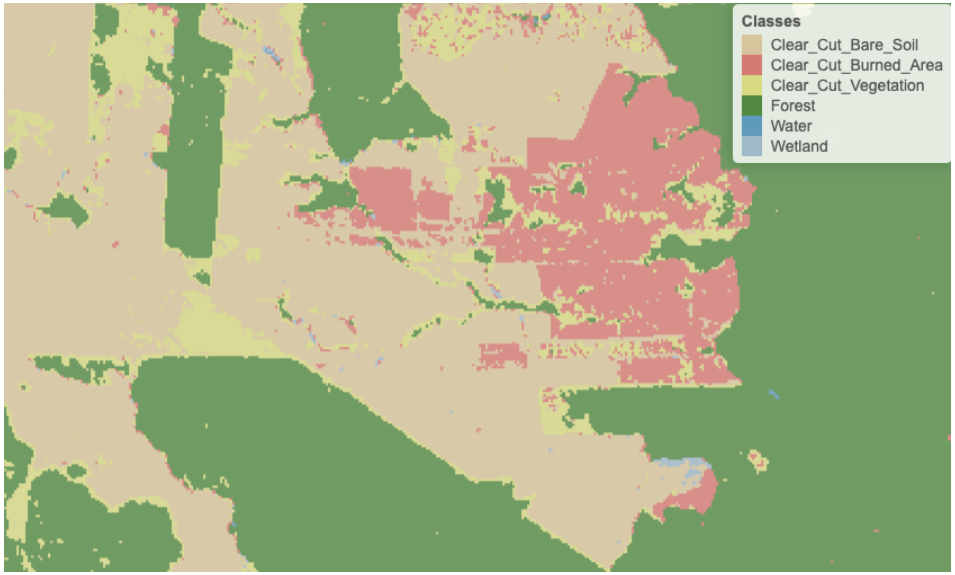
Are comparisons with previous work clear and explicit?

Does the paper describe something that has actually been implemented?

What should the reader learn from the paper?

```
self$conv_bn_relu1 <- .torch_conv1D_batch_norm_relu_dropout(
    input_dim    = n_bands,
    output_dim   = hidden_dims[[1L]],
    kernel_size  = kernel_sizes[[1L]],
    padding      = as.integer(kernel_sizes[[1L]] %/% 2L),
    dropout_rate = dropout_rates[[1L]]
)
# second module - 1D CNN
self$conv_bn_relu2 <- .torch_conv1D_batch_norm_relu_dropout(
    input_dim    = hidden_dims[[1L]],
    output_dim   = hidden_dims[[2L]],
    kernel_size  = kernel_sizes[[2L]],
    padding      = as.integer(kernel_sizes[[2L]] %/% 2L),
    dropout_rate = dropout_rates[[2L]]
)
# third module - 1D CNN
self$conv_bn_relu3 <- .torch_conv1D_batch_norm_relu_dropout(
    input_dim    = hidden_dims[[2L]],
    output_dim   = hidden_dims[[3L]],
    kernel_size  = kernel_sizes[[3L]],
    padding      = as.integer(kernel_sizes[[3L]] %/% 2L),
    dropout_rate = dropout_rates[[3L]]
)
# flatten 3D tensor to 2D tensor
self$flatten <- torch::nn_flatten()
# create a dense tensor
self$dense <- .torch_linear_batch_norm_relu_dropout(
    input_dim    = hidden_dims[[3L]] * n_times,
    output_dim   = dense_layer_nodes,
    dropout_rate = dense_layer_dropout_rate
)
```

Bayesian Inference for Post-Processing of Remote-Sensing Image Classification



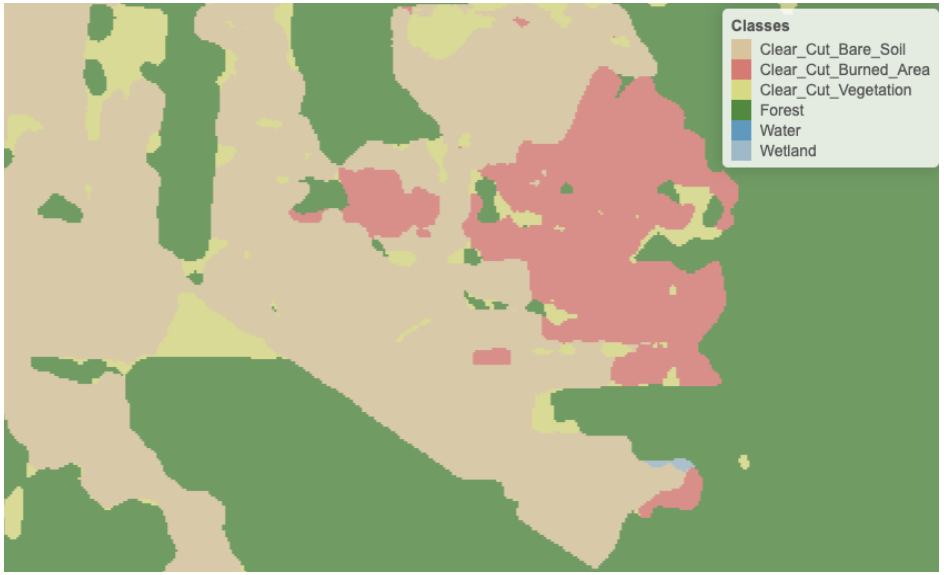


Table 3. Overall accuracy for points with high entropy values for their estimated probabilities.

Method	Accuracy
Original map	18.8%
Bayesian smoothing	68.3%
Gaussian smoothing	16.2%
Bilateral smoothing	19.1%

Bayesian Inference for Post-Processing of Remote-Sensing Image Classification

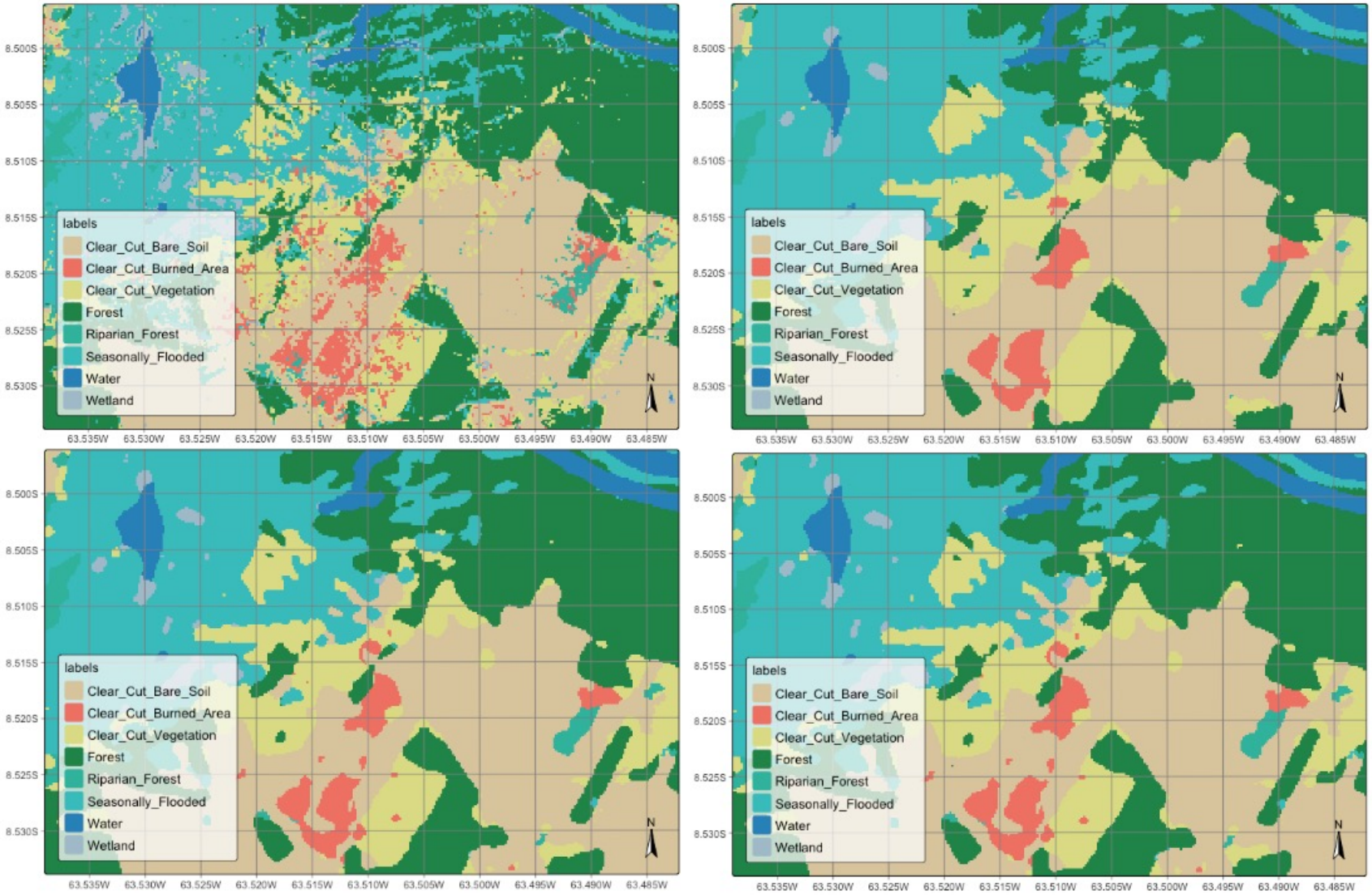
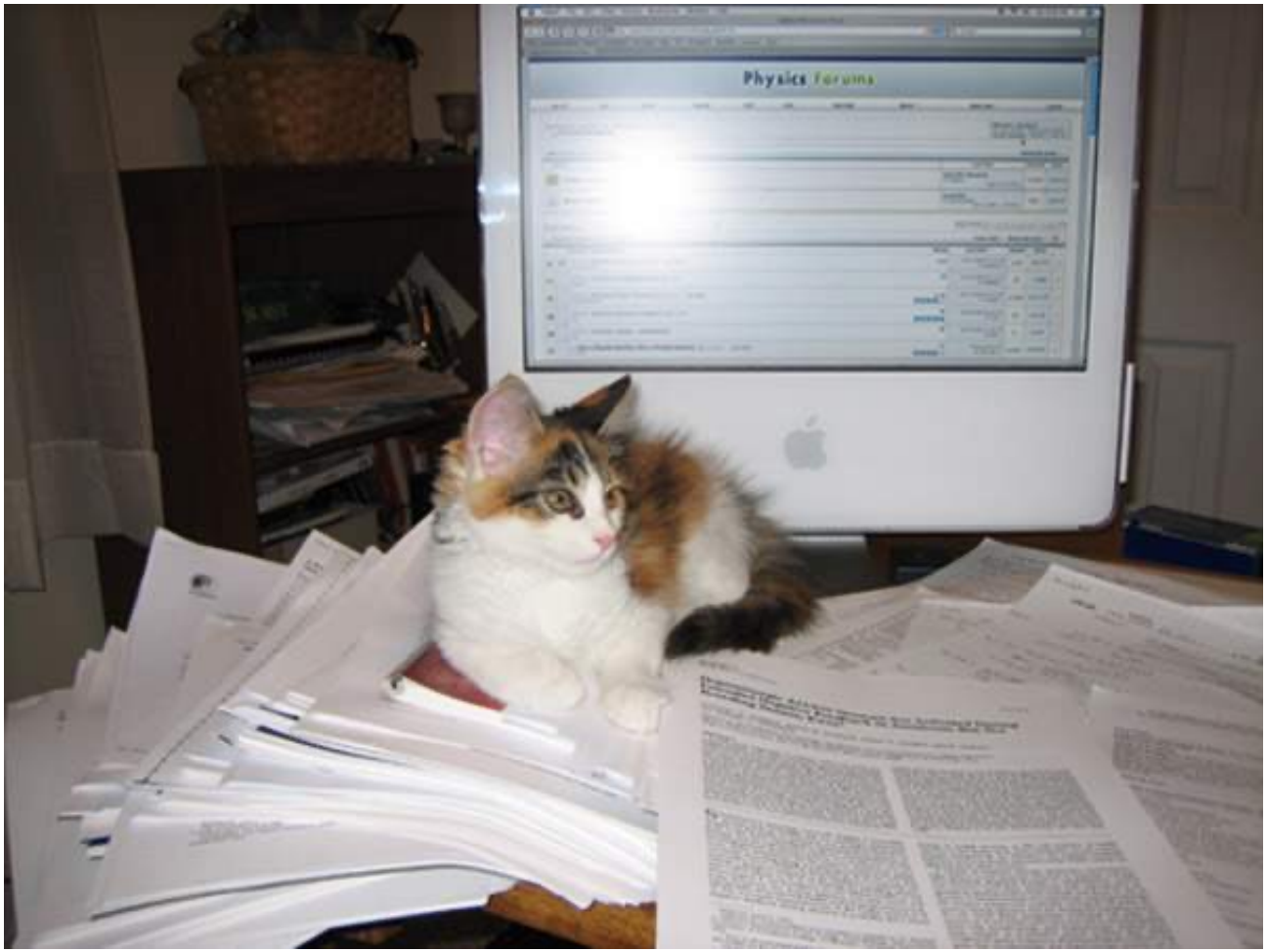


Figure 4. Detail of small area of classified map without smoothing (**top left**) and with Bayesian (**top right**), Gaussian (**bottom right**), and bilateral smoothing.

Informal research networks

Congresses, workshops, short schools, internships

Starting your research



Write a review paper (5-10 pgs) on the "state-of-the-art" in your research area

What is the state of the art?

Knowledge accumulated by your community

What is the state of the art?

State-of-the-art = Classics + recent advances

Google Scholar BETA

Web Images Video News Maps **more »**

deforestation tropical forests Search

Advanced
Scholar Pr
Scholar He

Scholar **All articles - Recent articles** Results **1** - **10** of about **46,10**

Tropical Deforestation and Habitat Fragmentation in the Amazon: Satellite Da
D Skole, C Tucker - Science, 1993 - sciencemag.org
... 1). In recent years, much attention has focused on **tropical forests**, where as much as 50% of the original extent may have been lost to **deforestation** in the ...
Cited by 746 - Related Articles - Web Search - Import into EndNote - BL Direct

Determination of **Deforestation** Rates of the World's Humid **Tropical Forests**
F Achard, HD Eva, HJ Stibig, P Mayaux, J Gallego, ... - Science, 2002 - sciencemag.org
... Reports. Determination of **Deforestation** Rates of the World's Humid **Tropical Forests**. Frédéric Achard, 1 * Hugh D. Eva, 1 Hans-Jürgen ...
Cited by 392 - Related Articles - Web Search - Import into EndNote - BL Direct

Proximate Causes and Underlying Driving Forces of **Tropical Deforestation** -
HJ GEIST, EF LAMBIN - BioScience, 2002 - bioone.org
... **Deforestation** of **Tropical** Rain **Forests**: Economic Causes and Impact on Developmer
Tropical forests: The main **deforestation** fronts. Environmental Conservation. ...
Cited by 241 - Related Articles - Web Search - Import into EndNote - BL Direct

Google Scholar

causes of tropical deforestation

Articles

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Custom range...

Sort by relevance
Sort by date

Any language
Search English pages

Any type
Review articles

include patents
include citations

Create alert

The **causes** of **tropical deforestation** in Ecuador: a statistical analysis
D Southgate, R Sierra, L Brown - World Development, 1991 - Elsevier
… strength of the various forces responsible for **deforestation**. That kind of … Our study of the **causes** of **tropical deforestation** was … Next, a model explaining **tropical deforestation** is presented. …
Save Cite Cited by 357 Related articles All 9 versions

[BOOK] The **causes** of **tropical deforestation**: the economic and statistical analysis of factors giving rise to the loss of the **tropical** forests
K Brown, DW Pearce - 1994 - books.google.com
… An underlying assumption of this volume, and the main reason why the **causes** of **tropical deforestation** is researched, is that **deforestation** has negative environmental impacts and …
Save Cite Cited by 473 Related articles All 9 versions

[HTML] Can economic **causes** of **tropical deforestation** be identified at a global level?
SS Scrieciu - Ecological Economics, 2007 - Elsevier
… **tropical deforestation**, … **deforestation** at the global level need to be carefully scrutinised and checked to ensure that they meet standard statistical tests. Second, **tropical deforestation** …
Save Cite Cited by 173 Related articles All 12 versions

Population, development and **tropical deforestation**: a cross-national study
T Rudel - The **causes** of **tropical deforestation**, 2023 - taylorfrancis.com
… partial tests of the adequacy of the population-growth and capital-availability explanations for **deforestation**, it should enhance our understanding ofthe **causes** of **tropical deforestation**. …
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SEMANTIC SCHOLAR

causes of tropical deforestation

About 1,970 results for "causes of tropical deforestation"

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences

Proximate Causes and Underlying Driving Forces of Tropical Deforestation

H. Geist E. Lambin Environmental Science, Economics · 1 February 2002

TLDR **Tropical deforestation** is driven by identifiable regional patterns **of** causal factor synergies, **of** which the most prominent are economic factors, institutions, national policies, and remote influences driving agricultural expansion, wood extraction, and infrastructure extension (at the proximate level). Expand

3,327 PDF BioOne Save Cite

Rethinking the causes of deforestation: lessons from economic models.

A. Angelsen D. Kaimowitz Environmental Science, Economics · The World Bank Research Observer · 1 February 1999

TLDR This synthesis **of** the results **of** more than 140 economic models analyzing the **causes of tropical deforestation** brings into question many conventional hypotheses upon **deforestation**. Expand

1,088 PDF PubMed Save Cite

Rocky desertification in Southwest China: Impacts, causes, and restoration

Zhongcheng Jiang Y. Lian X. Qin Environmental Science, Geography · 1 May 2014

Abstract Rocky desertification, which is relatively less well known than desertification, refers to the processes and human activities that transform a karst area covered by vegetation and soil into... Expand

997 Publisher Save Cite

Consensus

causes of tropical deforestation 1 | References | Share

causes of tropical deforestation

Pro · 2 steps · 20 sources

Main causes of tropical deforestation: agricultural expansion, wood extraction, infrastructure, and global economic forces

Overview

Across the tropics, **agricultural expansion is the dominant direct cause of deforestation**, reinforced by wood extraction (logging and fuelwood), infrastructure, mining, and large-scale land deals, all shaped by population, markets, governance, and policy.

Direct (proximate) causes

Land-use level drivers

Direct driver	Typical forms / examples	Citations
Agricultural expansion	Commercial and subsistence crops; cattle; tree plantations; oil palm, soy, rubber, cocoa, coffee, rice, cassava, maize	12 2 9 5 11 20
Wood extraction	Industrial logging, fuelwood,	9 1 19

Ask a follow up...

References / causes of tropical deforestation

Results

1 **Forest destruction in tropical Asia**

KEY TAKEAWAY · Tropical Asia is experiencing severe forest loss, with expanding human populations, industrial drivers, and endemic corruption contributing to the destruction.

HIGHLY CITED

2024 · 186 citations · W. Laurance · *Current Science*

2 **Drivers of deforestation and degradation for 28 tropical conservation landscapes**

KEY TAKEAWAY · Commercial and subsistence agriculture are the main drivers of deforestation in tropical conservation landscapes, with rice, rubber, cassava, and m being the most frequently cited crops.

OBSERVATIONAL STUDY HIGHLY CITED

2020 · 87 citations · H. Jayathilake et al.

3 **Analysis of the Social-Ecological Causes of Deforestation and Forest Degradation in Ghana: Application of the DPSIR Framework**

KEY TAKEAWAY · Ghana's deforestation and forest degradation are driven by agriculture expansion, cocoa farming expansion, illegal logging, illegal mining, population growth, and policy failures, despite numerous forest policies and interventions.

LITERATURE REVIEW HIGHLY CITED

Gemini

What are the seven most influential scientific papers about the causes of tropical deforestation and what are their assessments?

1. Geist, H. J., & Lambin, E. F. (2002). Proximate Causes and Underlying Driving Forces of Tropical Deforestation. *BioScience*.
2. Rudel, T. K., et al. (2009). Changing Drivers of Deforestation and New Opportunities for Conservation. *Conservation Biology.*
3. DeFries, R. S., et al. (2010). Deforestation Driven by Urban Population Growth and Agricultural Trade in the Twenty-First Century. *Nature Geoscience.*
4. Gibbs, H. K., et al. (2010). Tropical Forests Were the Primary Sources of New Agricultural Land in the 1980s and 1990s. *PNAS*.
5. Hosonuma, N., et al. (2012). An Assessment of Deforestation and Forest Degradation Drivers in Developing Countries. *Environmental Research Letters.*
6. Hansen, M. C., et al. (2013). High-Resolution Global Maps of 21st-Century Forest Cover Change. *Science*.
7. Nepstad, D., et al. (2014). Slowing Amazon Deforestation Through Public Policy and Interventions in Beef and Soy Supply Chains. *Science*.

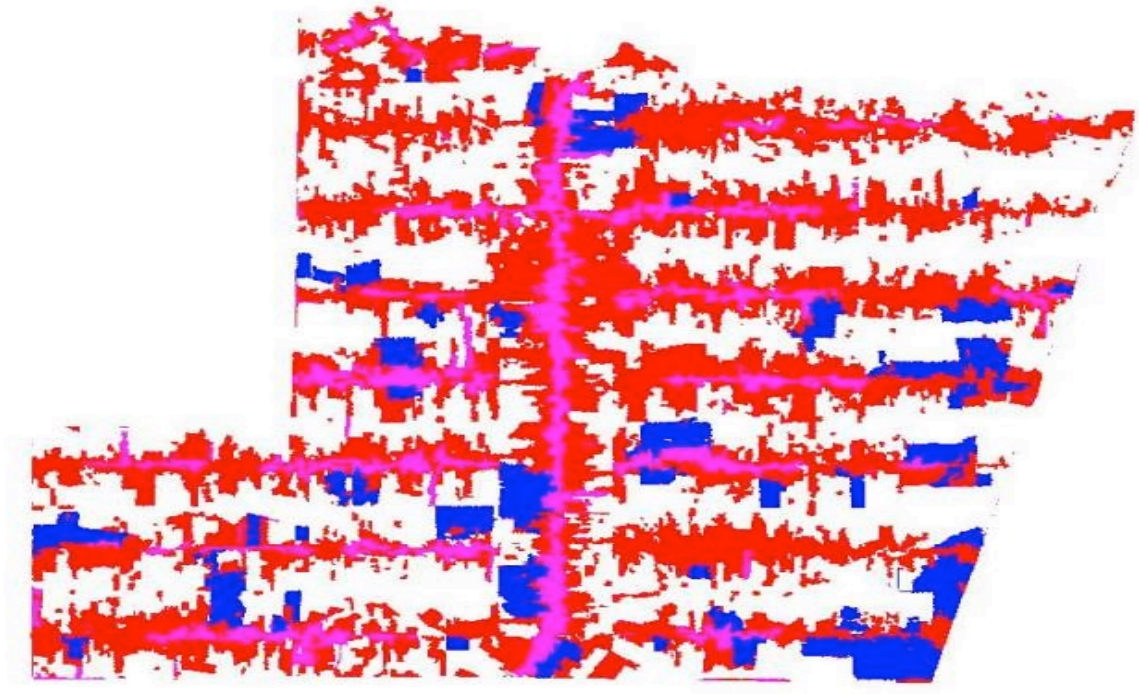
From general questions to scientific problems

"I'm going to study ___(topic)_____

"because I want to discover__(question)_____"

"to understand ___(relevance)_____

Spatial patterns of deforestation



I'm going to study methods to classify spatial patterns of deforestation... (topic)

because I want to figure out how to associate spatial patterns of deforestation with the different land uses... (question)

to understand who is deforesting the Amazon (relevance).

Amazonian manatee

I will study the spatial distribution of aquatic macrophytes consumed by the Amazonian manatee in the Middle Solimões region

Because I want to find out how the distribution of the main genera of macrophytes consumed by the manatee varies throughout the annual flood cycle in the region.

to understand how habitat structure and availability of food resources for manatees vary throughout the year.

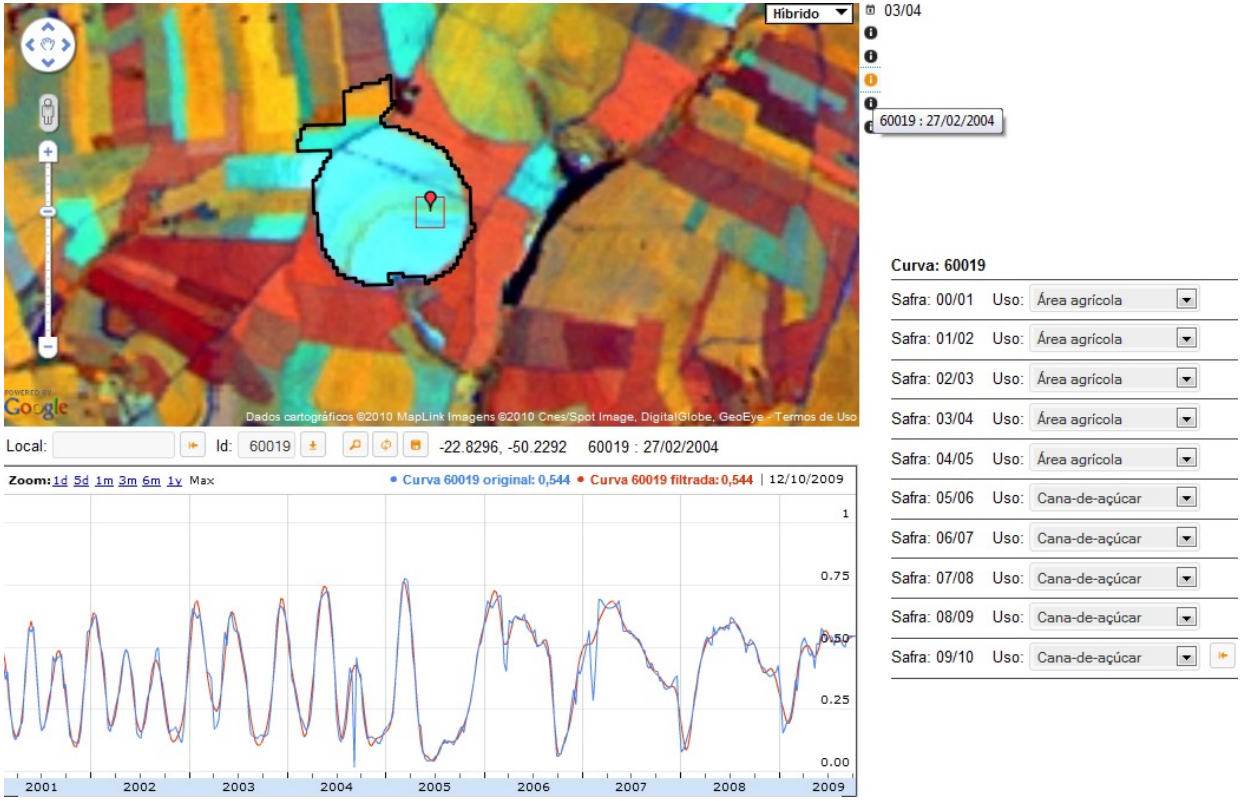
Dispersion of lobster larvae

I will study the dispersal of lobster larvae in the tropical Atlantic.

Because I want to find out if there is an interdependence between fish stocks.

To understand how it would be possible to establish legislation for fisheries resources that are exploited by more than one country

Land Use Change with Satellite Image Time Series

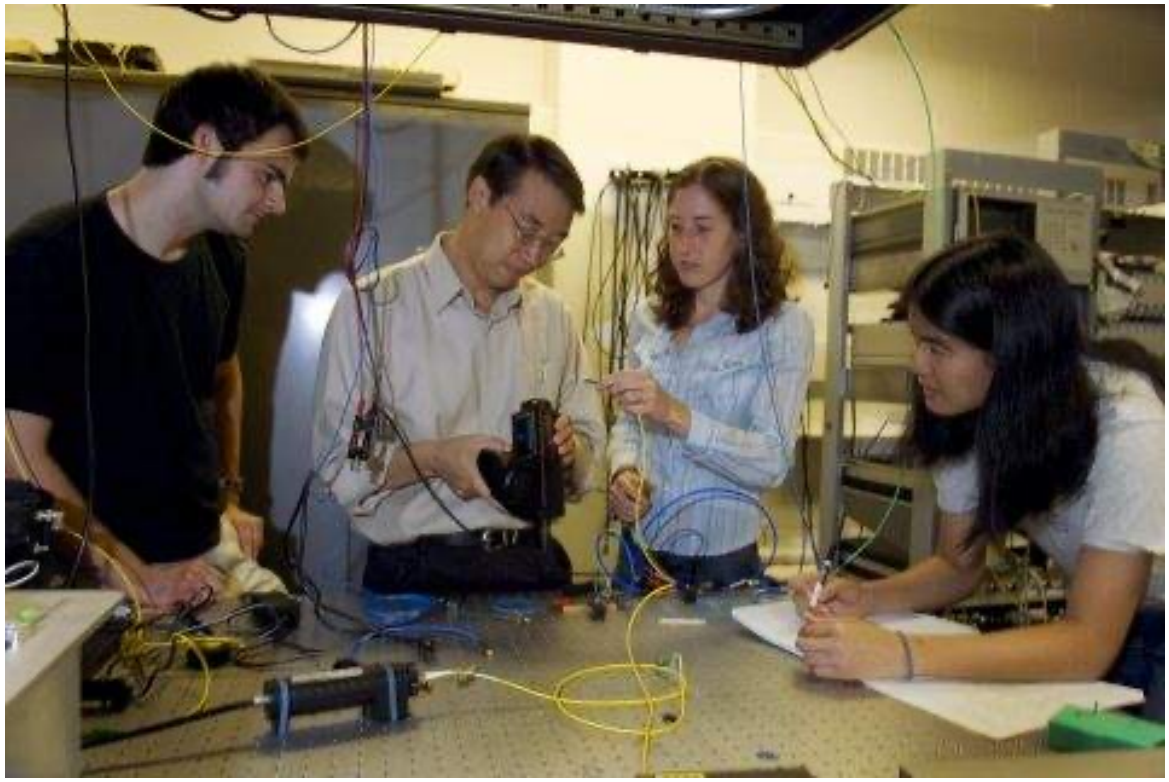


I will study how to extract information about land use transitions with time series from satellite imagery

Because I want to produce land use transition maps for Brazil

To understand what transformations have taken place in the Brazilian countryside since the beginning of the 21st century.

Preparation of the Thesis Proposal

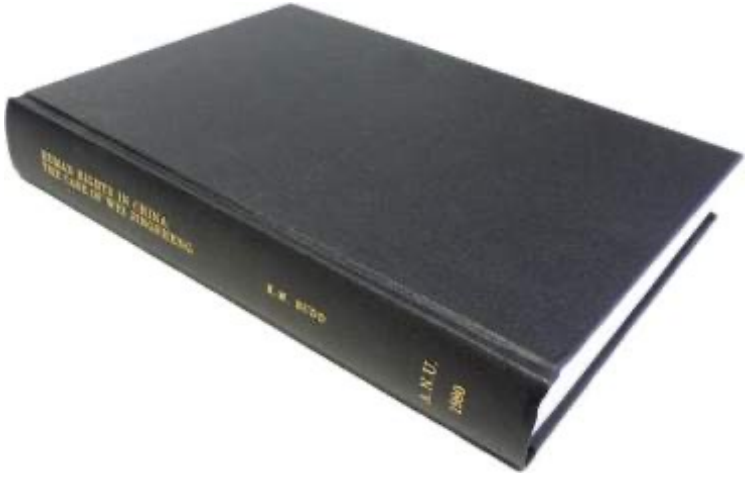


Join a support group

Present your ideas to your colleagues, your advisor and (if you can) at a scientific congress

Thesis Development

Submit scientific articles with results as soon as possible



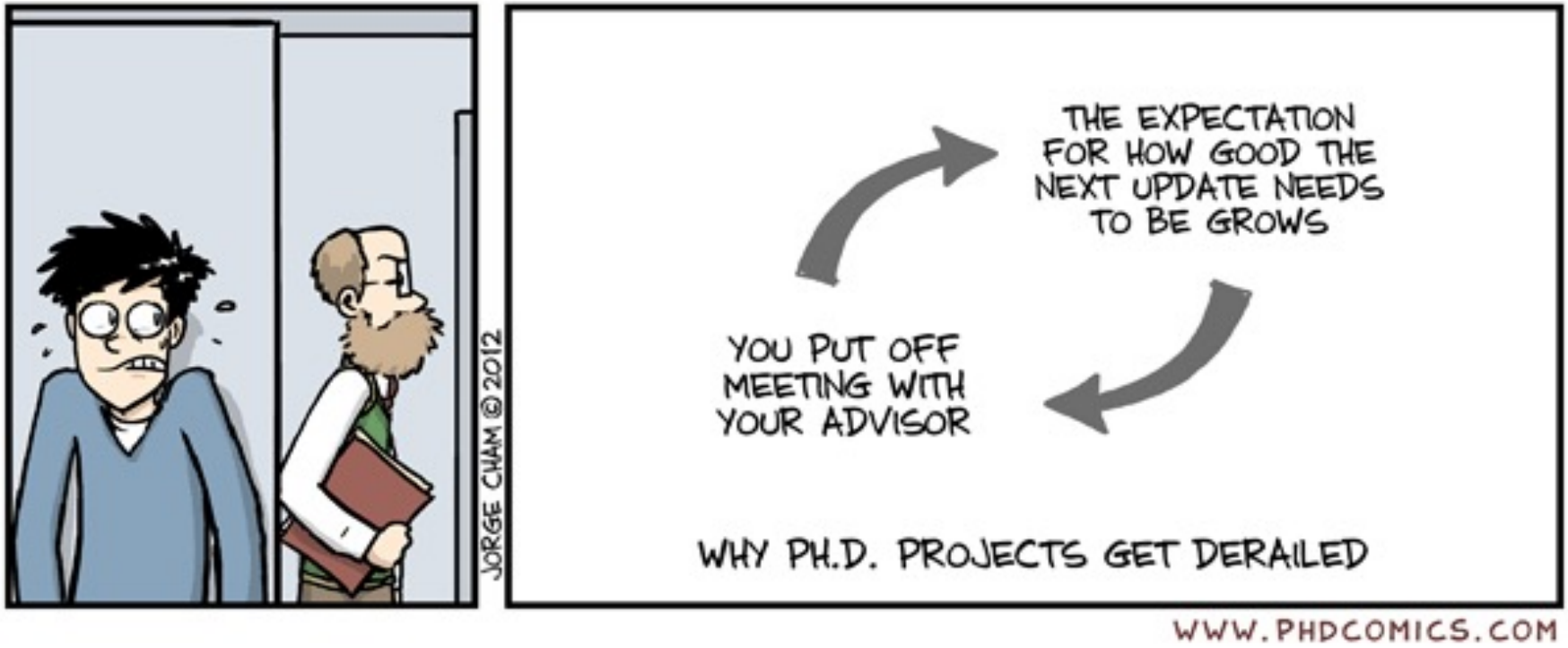
Compose the thesis from the articles

The basic principle of graduate studies

www.jamesyang.com

The thesis is yours, the advisor just shows you ways (and sometimes not even that...)

How to manage your advisor



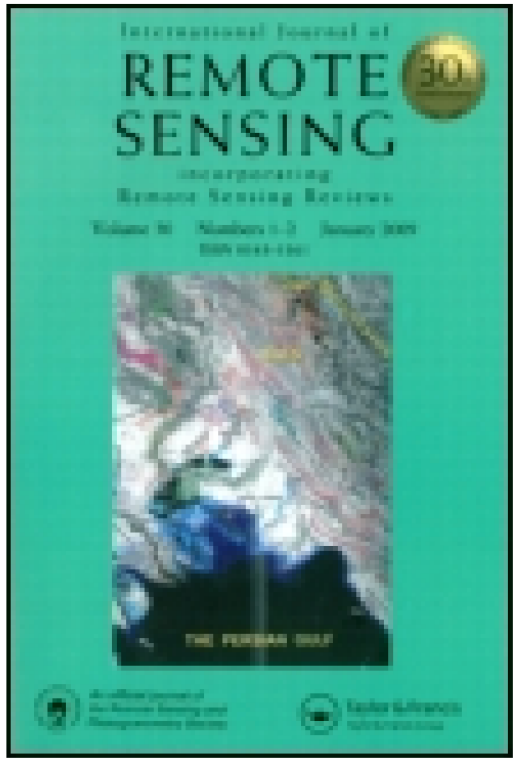
Hold regular meetings with your advisor (at least once a month)

Prepare documents for your advisor

The documents are for you to confirm your progress, not to receive praise from the advisor

How to manage your anxiety?

The story of a paper



International Journal of Remote Sensing

Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/tres20

Remote-sensing image mining: detecting agents of land-use change in tropical forest areas

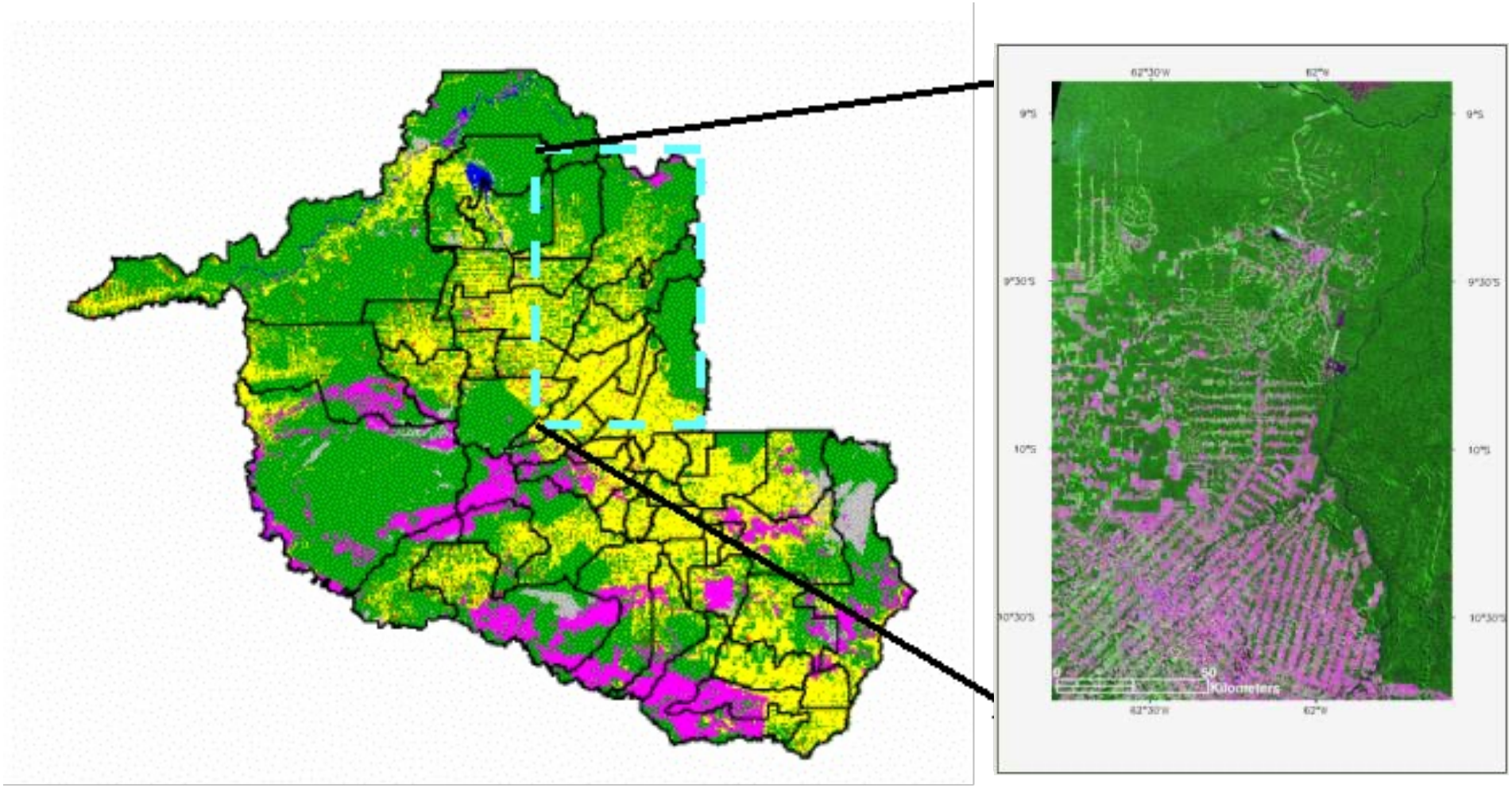
Marcelino Pereira dos Santos Silva [a b] , Gilberto Câmara [b] , Maria Isabel Sobral Escada [a] & Ricardo Cartaxo Modesto de Souza [b]

[a] UERN—Rio Grande do Norte State University , BR 110, km 46, 59625-620, Mossoró, RN, Brazil

[b] INPE—National Institute for Space Research , PO Box 515, 12201-097, São José dos Campos, SP, Brazil

Published online: 23 Jul 2008.

"Mining of Change Patterns in Remote Sensing Images" (PhD thesis by Marcelino Silva).

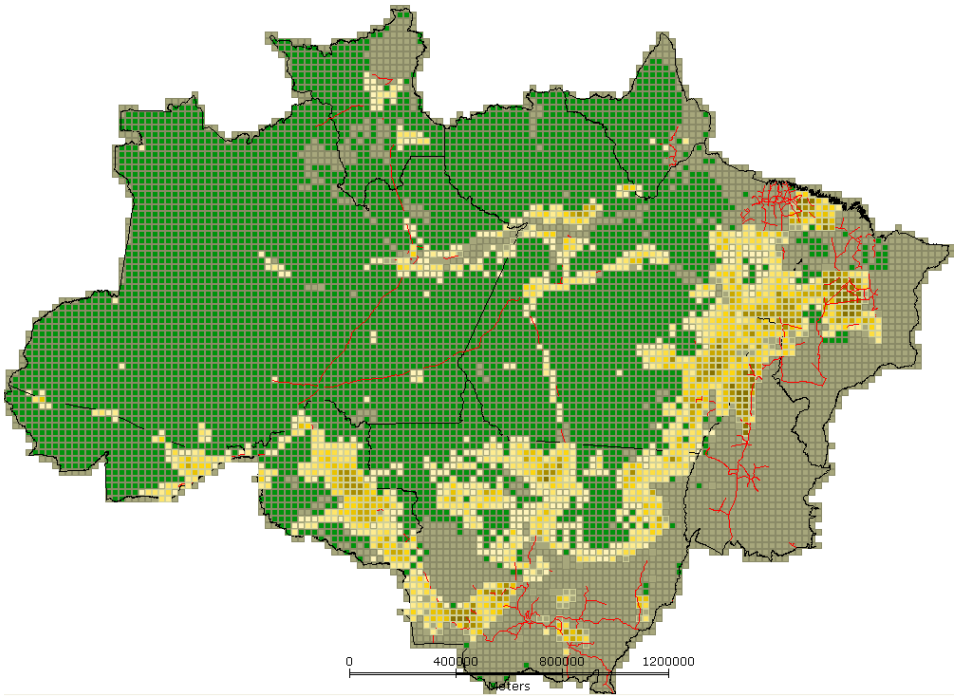


Escada, 2003.

Prodes (INPE, 2000)

TM/Landsat, 5, 4, 3 (2000)

Motivation

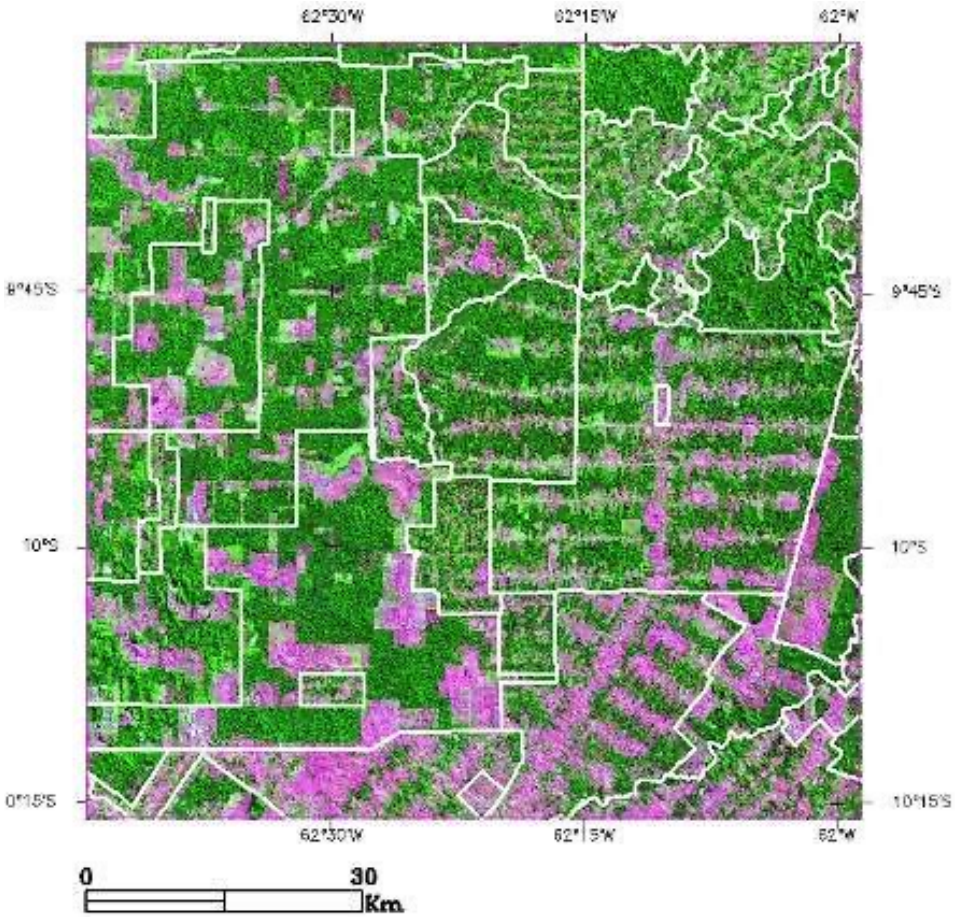


"We want to know who is deforesting the Amazon”

"The data we have are from satellite images”

"Can you relate the patterns measured in the images to the profile of those who deforested?

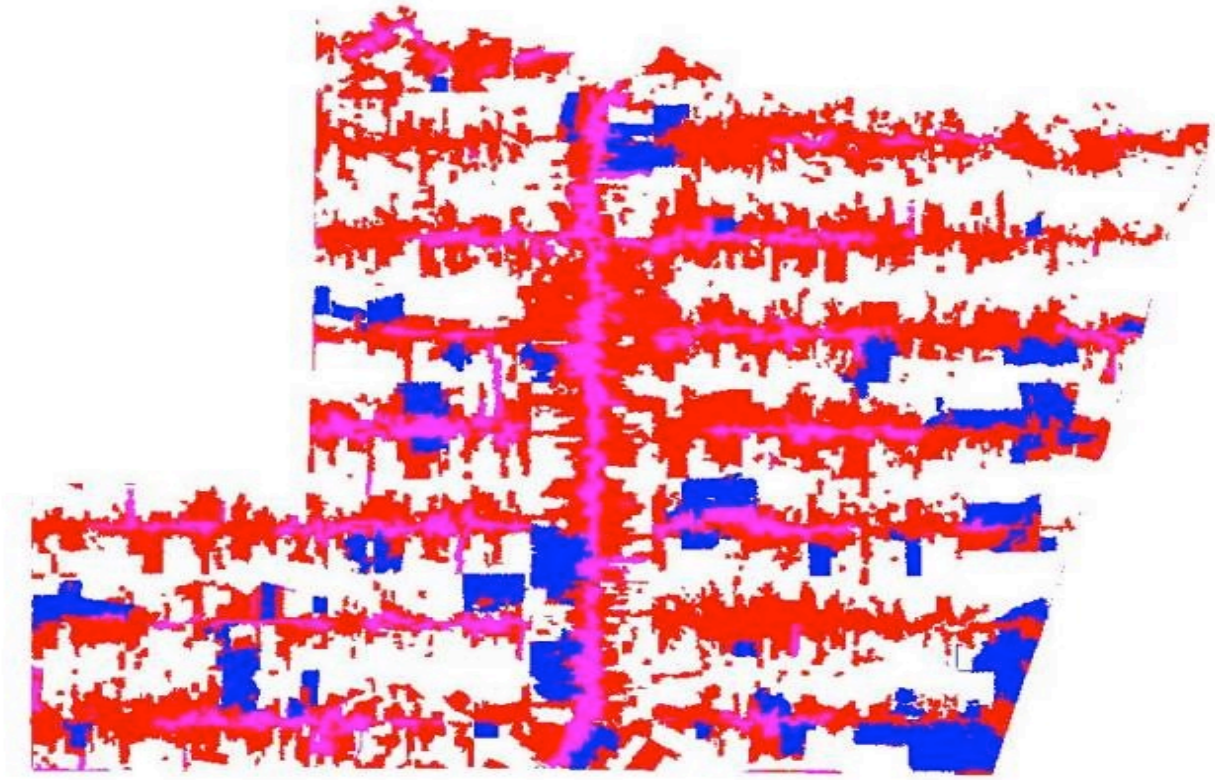
From questions to hypothesis



Question:
How can we use spatial patterns measured by images to infer who is deforesting?

Hypothesis:
Each occupation has a typical shape, detectable in the image. If we associate form with agents, we will know who deforested.

How to use spatial patterns measured by images to infer who is deforesting?



I am studying methods to classify spatial patterns of deforestation

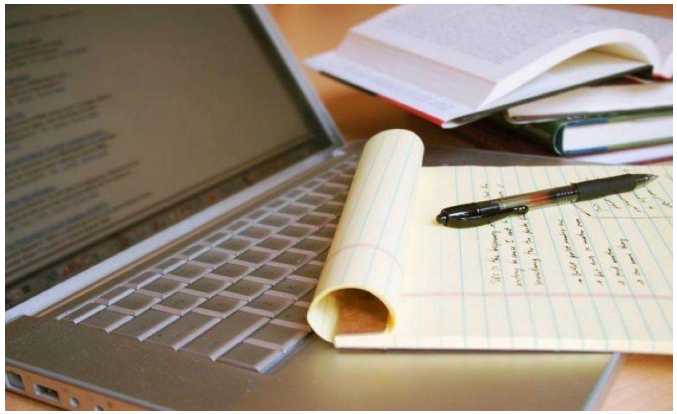
Because I want to find out how to associate spatial patterns of deforestation with land-use change agents

To understand who is deforesting the Amazon

What is the spatial pattern of each agent?

Spatial Patterns in Rondônia

"Mining Patterns of Change in Remote Sensing Imagery"



GeoInfo 2004
Acceptance Rate – 40%
(rejected)

↓

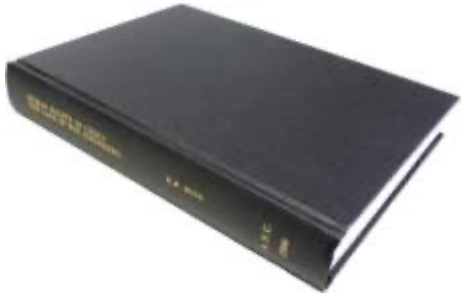
SSTD 2005
Acceptance Rate – 25%
(rejected)

↓

IEEE Data Mining 2005
Acceptance Rate – 10%
(accepted)

↓

Int Journal Remote Sensing
(accepted)

↓

Thesis – expanded version of the IJRS paper